

QP CODE: 4220100915



Reg No : .....

Name : .....

**M.C.A. DEGREE EXAMINATION, JANUARY 2022**

**First Semester**

**Core - MCACT105 - DATABASE TECHNOLOGY AND NOSQL**

2020 Admission Onwards

2C3C51B1

Time: 3 Hours

Maximum: 75 Marks

**Part A**

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Discuss the advantages of DBMS.
2. Illustrate schema and instance with examples.
3. Illustrate conceptual schema with example.
4. Explain the concept of relational model with a suitable example.
5. Represent the way in which Primary key, Candidate key and Super key differs in its working?
6. List out the features of Weak entities?
7. What are integrity Constraints used in SQL
8. Explain Functional dependancy with example
9. What is the significance of Transaction Log File ?
10. What are Dead Locks?
11. Write short notes on Schemaless Database.
12. Illustrate Database sharding.

(10×3=30 marks)

**Part B**

*Answer **all** questions*

*Each question carries **9** marks*

13. a) Describe the main characteristics of the database approach in contrast with the file sytem oriented approach

OR





- b) Design an ER diagram for the NHL(National Hockey League) database with these details:- the NHL has many teams, each team has a name, a city, a coach, a captain, and a set of players, each player belongs to only one team, each player has a name, a position (such as left wing or goalie), a team captain is also a player, a game is played between two teams (referred to as host\_team and guest\_team) and has a date and a score.
14. a) Write about Primary key, Foreign key and other Integrity constraints in a relation with suitable examples.

OR

- b) Explain how to translate the relationship sets with key constraints into relational table?
15. a) Consider the employee database
- Employee** (Employee\_name, street, city)
- Works** (Employee\_name, Company\_name, salary)
- Company** (company\_name, city)
- Manager** (employee\_name, manager\_name)
- Give SQL expression for the following

- i. *Find the name and cities of residence of all employees who work for first Bank Corporation.*
- ii. *Find the name, street and cities of residence of all employees who work for First Bank Corporation and earn more than \$ 10,000.*
- iii. *Find all employees in the database who live in the same cities and on the same streets.*
- iv. *Find the employees whose salary is greater than ten thousand.*

OR

- b) Compare BCNF and 3NF with example.
16. a) Explain the need for concurrency control. How the issues of concurrent execution is solved.

OR

- b) Explain how crash recovery is performed using ARIES.
17. a) Illustrate Document store for Customer details and their orders.

OR

- b) Discuss Master slave Replication in NoSQL.

(5×9=45 marks)

